# Emotion Detection from Speech Audio

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## APEX Affirmation

I affirm that this written product represents my own original work, completed under the supervision of my faculty mentor, and that all sources have been properly acknowledged in accordance with academic integrity standards. This document fulfills the written product requirement of the Academic Program for Excellence (APEX) at the University of Massachusetts Dartmouth.

# Abstract

Speech Emotion Recognition (SER) is the automated classification of a speaker's emotional state from audio signals, a problem with applications spanning mental health monitoring, human-computer interaction, and clinical diagnostics. Despite decades of research, SER remains a difficult open problem, in part because the majority of published work evaluates models under speaker-dependent conditions, allowing models to learn voice identity rather than emotional content and artificially inflating reported accuracy by 10 - 15 percentage points relative to more rigorous evaluations.

This project designed and implemented a complete, speaker-independent SER pipeline using two publicly available datasets: the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS; Livingstone & Russo, 2018) and the Crowd-Sourced Emotional Multimodal Actors Dataset (CREMA-D; Cao et al., 2014), yielding 7,035 audio clips across 115 speakers. All train, validation, and test splits were partitioned strictly by actor identity, with a hard programmatic assertion preventing actor overlap between splits.

Six model architectures were trained and compared: a Support Vector Machine (SVM) on 260-dimensional handcrafted scalar features, a CNN + Bidirectional LSTM with attention on Mel spectrograms, an EfficientNet-B0 on multi-channel spectrogram images, a ResNet-18 dual-input model combining spectral and waveform features, a Multi-Feature CNN + BiLSTM fusing all acoustic feature types jointly, and a fine-tuned Wav2Vec2-base transformer pre-trained on 960 hours of unlabeled speech. Test accuracies ranged from 52.6% (SVM) to 75.3% (Wav2Vec2, Macro F1: 0.749). The best-performing model was deployed as a live interactive web application accessible at https://huggingface.co/spaces/T22S/ser-emotion-demo. Code is available at https://github.com/TanmaySonawane/Improved_Emotion_Detection.

These results demonstrate that transfer learning from large-scale self-supervised speech pretraining substantially outperforms architectures trained from scratch on limited labeled data, and that speaker-independent evaluation produces meaningfully lower and more honest accuracy figures than most published benchmarks report.

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# 1. Purpose of the APEX

The central question this project addressed is: *Can a deep learning system reliably identify human emotion from speech audio when evaluated under strict speaker-independent conditions?*

This question has both a scientific dimension and a methodological one. On the scientific side, the goal was to determine which class of model architecture: classical machine learning, convolutional-recurrent networks trained from scratch, transfer learning from vision models, or transformer-based self-supervised pretraining, achieves the highest generalization accuracy on a five-class emotion recognition task (angry, fear, happy, neutral, sad) when trained and evaluated on audio from speakers entirely absent from the training set.

On the methodological side, a central motivation was to address a pervasive flaw in the SER literature: the widespread use of speaker-dependent evaluation protocols, which allow the same actors to appear in both training and test data. This practice allows models to learn voice identity as a proxy for emotion, producing inflated accuracy estimates that do not reflect real-world performance on unseen speakers. This project enforced strict speaker independence at every stage, a constraint that is technically straightforward to implement but absent from a large fraction of published work.

Beyond the research question, this project was designed as an end-to-end engineering exercise: beginning from raw audio files, building a complete feature extraction and augmentation pipeline, training and systematically comparing multiple model architectures, evaluating results rigorously, and deploying the final system as a live web application available to anyone. This reflects a professional standard of completeness that goes beyond a single model evaluation.

The project was completed as a solo endeavor under the mentorship of Dr. Alfa Heryudono of the UMass Dartmouth Department of Mathematics, whose guidance shaped the statistical rigor of the evaluation methodology and the systematic multi-architecture comparison design.

# 2. Scholarly Background and Literature Review

## 2.1 Speech Emotion Recognition: Overview and Applications

Speech Emotion Recognition is a subfield of affective computing concerned with automatically inferring the emotional state of a speaker from acoustic properties of their speech signal. The problem has attracted sustained research interest since the early 2000s, with applications including clinical mental health monitoring (detecting depression or anxiety from patient speech), human-robot interaction (enabling machines to respond appropriately to user frustration or distress), and call center analytics (flagging customer dissatisfaction in real time). Schuller (2018) provides a comprehensive two-decade review of the field, noting that while significant progress has been made, generalization to naturalistic, spontaneous speech remains an unresolved challenge.

The human perception of emotion from speech relies on a combination of acoustic cues. Pitch (fundamental frequency) tends to rise during high-arousal states such as anger and fear; speech rate accelerates under excitement; voice quality (breathiness, roughness) shifts with emotional intensity; and energy distribution across frequency bands reflects emotional coloring that trained listeners reliably exploit. Automatic systems must learn to extract and weight these cues from numerical representations of audio waveforms.

## 2.2 Feature Representations for Speech

A core challenge in SER is selecting acoustic features that capture emotionally relevant information while remaining invariant to speaker identity, recording environment, and speaking style. The research community has converged on several standard representations.

**Mel-Frequency Cepstral Coefficients (MFCCs)** are among the most widely used features in speech processing. MFCCs compress the spectral envelope of a speech frame into a small set of coefficients (typically 13 - 40) computed on the Mel frequency scale, which approximates the nonlinear frequency resolution of the human auditory system. Including the first and second temporal derivatives (delta and delta-delta) adds rate-of-change information, making the feature sensitive to transitions between phonemes and emotional states (Logan, 2000; McFee et al., 2015).

**Mel spectrograms** represent audio as a two-dimensional time-frequency image on the

Mel scale, with time on one axis and Mel frequency bins on the other. Unlike MFCCs, which discard phase and compress frequency detail, Mel spectrograms retain richer frequency resolution and are well-suited to image-based deep learning architectures such as convolutional neural networks.

**Chroma features** represent the distribution of energy across the 12 pitch classes of the Western musical scale, capturing harmonic and tonal properties of speech. **Spectral contrast** measures the difference in energy between peaks and valleys in successive frequency sub-bands, quantifying the tonal versus noise-like character of the signal. Both have been used as supplementary features in SER systems.

### 2.3 Machine Learning Approaches to SER

**Classical machine learning.** Early SER systems relied on handcrafted feature vectors: statistical summaries (mean, standard deviation, kurtosis) of frame-level acoustic descriptors, fed into classifiers such as Gaussian Mixture Models (GMMs), Hidden Markov Models (HMMs), and Support Vector Machines (SVMs). SVMs in particular became a dominant baseline during the 2000s and 2010s, achieving competitive performance on speaker-dependent benchmarks while remaining interpretable and computationally lightweight (Schuller, 2018). However, handcrafted features cannot capture complex temporal patterns that span multiple seconds and require domain expertise to design.

**Convolutional and recurrent architectures.** The availability of larger labeled datasets and GPU computing made it feasible to train deep neural networks end-to-end on SER tasks. Convolutional Neural Networks (CNNs) applied to Mel spectrograms learn hierarchical frequency-domain patterns without manual feature engineering. Recurrent architectures, particularly Bidirectional Long Short-Term Memory networks (BiLSTMs) capture long-range temporal dependencies in speech, making them well-suited to detecting emotion patterns that evolve over the duration of an utterance. Combined CNN + BiLSTM architectures have achieved strong results on several benchmarks by first extracting local spectral features with convolutions and then modeling their temporal evolution with recurrence (Zhao et al., 2019).

**Transfer learning from vision.** A practical challenge in SER is data scarcity: even large emotional speech databases contain at most tens of thousands of clips, far fewer than the millions of examples used to train deep vision models. One approach to this problem is to treat Mel spectrograms as images and fine-tune ImageNet-pretrained architectures such as

ResNet (He et al., 2016) and EfficientNet (Tan & Le, 2019) on SER. This leverages rich low-level feature detectors learned from visual data, edge and texture detectors that generalize surprisingly well to spectrogram patterns, while requiring only the final classification layers to adapt to the audio domain (Hershey et al., 2017).

**Self-supervised learning and transformer models.** The most significant recent development in speech processing has been the emergence of self-supervised transformer models pretrained on massive quantities of unlabeled audio. Wav2Vec 2.0 (Baevski et al., 2020), developed by Meta AI Research, introduced a framework in which a CNN feature extractor converts raw waveforms into dense frame-level representations, which are then processed by a 12-layer transformer encoder trained to predict masked portions of the input. Crucially, this pretraining is performed on 960 hours of unlabeled speech from the LibriSpeech corpus (Panayotov et al., 2015), requiring no emotion labels. The resulting representations capture rich acoustic properties: prosody, phoneme boundaries, vocal effort, that are directly relevant to emotion recognition. Pepino et al. (2021) demonstrated that even shallow fine-tuning of Wav2Vec2-base embeddings on emotional speech data substantially outperforms models trained from scratch on equivalent labeled data, establishing self-supervised pretraining as the current state-of-the-art approach for SER under limited-data conditions.

### 2.4 The Speaker Independence Problem

A critical and underappreciated methodological issue in SER research is the distinction between speaker-dependent and speaker-independent evaluation. In speaker-dependent protocols, the same actors appear in training and test sets, only different utterances from the same speaker are separated. This is problematic because it allows models to learn speaker-specific vocal characteristics (timbre, pitch range, speaking style) as shortcuts for emotion prediction, rather than learning the underlying acoustic correlates of emotion that generalize across speakers. Schuller (2018) notes that speaker-dependent systems routinely report 90 - 95% accuracy on RAVDESS, while speaker-independent systems on the same dataset typically achieve 70 - 80%, a gap of 10 - 20 percentage points that reflects inflated metrics rather than genuine improvement in emotion understanding.

This project enforces strict speaker independence: the 24 RAVDESS actors and 91 CREMA-D actors were partitioned at the actor level (not the utterance level) into training, validation, and test splits, with a programmatic assertion that crashes execution if any actor identity appears in more than one split.

### 2.5 Datasets

**RAVDESS** (Livingstone & Russo, 2018) is a widely used emotional speech and song database recorded by 24 professional actors (12 male, 12 female) in studio conditions. Utterances are recorded at high audio quality in North American English, expressing eight emotions at two intensity levels. For this project, five emotions were retained (angry, fearful, happy, neutral, sad) and calm, disgust, and surprised were excluded to match the CREMA-D label set, yielding approximately 864 usable clips.

**CREMA-D** (Cao et al., 2014) is a larger, more diverse corpus of 7,442 clips from 91 actors (48 male, 43 female) representing a wide range of ages, ethnicities, and regional accents, recorded in variable acoustic conditions. Six emotions are labeled: anger, disgust, fear, happiness, neutral, and sadness. Disgust was excluded to match the five-class schema, yielding approximately 6,171 usable clips.

Together, the two datasets provide 7,035 clips across 115 unique speakers, a combination that improves both the size and demographic diversity of the training set compared to either dataset alone.

# 3. Methods

## 3.1 Datasets, Preprocessing, and Speaker-Independent Splitting

All audio files were resampled to 22,050 Hz and converted to mono. Each clip was standardized to exactly four seconds (88,200 samples) using center-padding with silence for shorter clips and center-cropping for longer clips. This ensures that all feature arrays have identical dimensions regardless of the original recording length.

Speaker-independent splits were constructed by partitioning actor IDs and not individual files, into training, validation, and test sets using a fixed random seed (42) applied independently to each dataset. This preserves proportional representation of both RAVDESS and CREMA-D actors across all three splits. The resulting split is: 70% train (80 actors, approximately 4,924 clips before augmentation), 15% validation (17 actors, approximately 1,056 clips), and 15% test (18 actors, approximately 1,055 clips). A programmatic assertion runs after splitting and raises a fatal error if any actor identity appears in more than one split, making data leakage detectable and impossible to overlook.

## 3.2 Feature Extraction

Six feature representations were computed from each audio clip using the librosa library (McFee et al., 2015):

**Table 1:** Feature representations extracted from each audio clip.

| Feature | Shape | Description |
|---|---|---|
| Mel spectrogram | $(128, 173)$ | Energy at each of 128 Mel-frequency bins over 173 time frames |
| MFCC + $\Delta$ + $\Delta\Delta$ | $(40, 173, 3)$ | 40 cepstral coefficients plus first and second temporal derivatives |
| Chroma | $(12, 173)$ | Energy distribution across 12 pitch classes over time |
| Spectral contrast | $(7, 173)$ | Peak-to-valley energy ratio in 7 frequency sub-bands over time |
| Raw waveform | $(88{,}200, )$ | Raw audio samples at 22,050 Hz |
| Scalar features | $(260, )$ | Statistical summaries (mean, std.) of all frame-level features above |

MFCCs were normalized per-row per-channel to zero mean and unit standard deviation to prevent energy-scale differences from dominating the representation. All features were

saved as NumPy arrays on disk, enabling fast data loading during training without repeated recomputation.

### 3.3 Data Augmentation

The training set only (never validation or test) was expanded five-fold through offline augmentation. For each original training clip, four augmented variants were generated by modifying the raw audio waveform before recomputing all six feature types:

1. **Gaussian noise addition:** Random Gaussian noise scaled to 0.5% of the clip's maximum absolute amplitude was added to each sample, simulating microphone noise or ambient room sound.
2. **Pitch shifting:** The pitch was shifted by $\pm 1$ semitone using the phase vocoder algorithm (librosa's `pitch_shift`), which applies a Short-Time Fourier Transform (STFT), multiplies frequency components by the semitone ratio ($2^{1/12} \approx 1.059$) or ($2^{-1/12} \approx 0.944$), and reconstructs the time-domain signal via inverse STFT.
3. **Time stretching:** The audio was compressed (rate = 1.1, approximately 10% faster) or expanded (rate = 0.9, approximately 10% slower) without changing pitch using the phase vocoder's time-scale modification. The resulting clip was re-padded or re-cropped to four seconds.
4. **Volume scaling:** The amplitude of each sample was multiplied by a constant gain factor (1.3× or 0.7×), simulating variation in recording distance and microphone sensitivity.

A programmatic assertion verified after augmentation that no augmented rows appeared in the validation or test manifests, ensuring augmentation introduced no evaluation leakage. The final training set contained approximately 24,500 clips.

Online augmentation (SpecAugment; Park et al., 2019) was additionally applied during training for CNN-based models: random time and frequency masks were applied to Mel spectrograms within each training batch, further improving regularization without additional disk storage.

### 3.4 Model Architectures

Six architectures of increasing complexity were trained and compared.

**Model 1 :- SVM and Random Forest Baseline.** A 260-dimensional scalar feature vector (statistical summaries of all frame-level features) was computed for each clip and used to train a Support Vector Machine with a radial basis function kernel and a Random Forest with 200 trees using scikit-learn. Features were standardized with a StandardScaler fit exclusively on the training set. These models serve as interpretable baselines reflecting the classical SER approach.

**Model 2 :- CNN + Bidirectional LSTM with Attention.** A 128-channel Mel spectrogram ($128 \times 173$) was processed by four convolutional blocks, each including 2D convolution, batch normalization, ReLU activation, and Squeeze-Excitation channel recalibration (Hu et al., 2018). Pooling was applied along the frequency axis only, preserving the 173-frame time dimension. The resulting sequence was passed through a two-layer BiLSTM (hidden size 128 per direction), and a self-attention layer weighted the 173 time steps to emphasize emotionally salient moments. Spatial Dropout regularized temporal representations before the final linear classifier.

**Model 3 :- EfficientNet-B0 on Multi-Channel Spectrograms.** The Mel spectrogram, chroma features (resized from $12 \times 173$ to $128 \times 173$), and spectral contrast features (resized from $7 \times 173$ to $128 \times 173$) were stacked as a three-channel image (analogous to RGB) and resized to $224 \times 224$. An ImageNet-pretrained EfficientNet-B0 (Tan & Le, 2019) processed this image in a two-phase training regime: the backbone was frozen for ten epochs while only the classification head was trained, then the final three MBConv blocks were unfrozen and fine-tuned at a reduced learning rate of $5 \times 10^{-5}$.

**Model 4 :- ResNet-18 Dual-Input.** Two parallel input streams were combined. Stream A processed the Mel spectrogram through an ImageNet-pretrained ResNet-18 backbone (He et al., 2016), using a learned $1 \times 1$ convolution to expand from one channel to three. Stream B processed the raw waveform (downsampled to 8 kHz) through four 1D convolutional layers. The embeddings from both streams (512-dim and 256-dim, respectively) were concatenated and classified jointly by a two-layer head, allowing the model to jointly reason about spectral structure and temporal waveform properties.

**Model 5 :- Multi-Feature CNN + BiLSTM.** All per-frame feature arrays (Mel, flattened MFCC+$\Delta$+$\Delta\Delta$, chroma, spectral contrast) were stacked along the frequency axis into a

single $267 \times 173$ array, then processed by a five-block CNN + BiLSTM pipeline. This unified representation allows the model to learn cross-feature interactions, for example, the joint pattern of high-pitch Mel energy and rapidly changing MFCC deltas that characterizes angry speech rather than treating each feature independently. BiLSTM hidden size was increased to 256 per direction following recommendations from multi-feature SER literature (Zhao et al., 2019).

**Model 6 :- Wav2Vec2-base Fine-Tuned (Best Model).** The raw waveform (88,200 samples at 22,050 Hz) was resampled on-the-fly to 16,000 Hz (64,000 samples) inside the model's forward pass to match the sample rate on which `facebook/wav2vec2-base` was pretrained. The model architecture consists of a seven-layer CNN feature extractor that converts the waveform into approximately 200 frame-level representations, followed by a 12-layer transformer encoder with full self-attention across all frames. The bottom six transformer layers were frozen (preserving learned low-level acoustic representations), and the top six layers plus a two-layer classification head (Linear($768 \rightarrow 256$) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear($256 \rightarrow 5$)) were fine-tuned. Training used a batch size of 8 (constrained by GPU VRAM), learning rate of $1 \times 10^{-4}$ (one order of magnitude lower than scratch-trained models to prevent catastrophic forgetting), and CosineAnnealingWarmRestarts scheduling.

### 3.5 Training Procedure

All PyTorch models were trained on an NVIDIA GeForce RTX 3050 Ti (4 GB VRAM) using the following shared procedures:

- **Loss function:** Cross-entropy with class-balanced weights (inversely proportional to class frequency) and label smoothing of 0.1 to prevent overconfident predictions on a noisy, subjectively labeled task.
- **Optimizer:** AdamW with weight decay of $1 \times 10^{-4}$.
- **Automatic Mixed Precision (AMP):** All training loops used PyTorch's `autocast` and `GradScaler` to reduce VRAM usage and accelerate training without sacrificing numerical precision.
- **Early stopping:** Training was halted if validation accuracy did not improve for 12 - 15 consecutive epochs, with the best-performing checkpoint saved.

### 3.6 Ensemble and Deployment

An ensemble combining the outputs of all trained models was constructed by optimizing per-model weights using the Sequential Least Squares Programming (SLSQP) algorithm from `scipy.optimize.minimize`, maximizing validation accuracy subject to the constraint that weights sum to one and are non-negative. Test-Time Augmentation (TTA) averaged predictions over three augmented views of each test clip.

The best-performing model (Wav2Vec2) was deployed as a Gradio web application hosted on HuggingFace Spaces. The application accepts either microphone input or an uploaded audio file, preprocesses it to the four-second standardized format, runs inference through the fine-tuned model, and returns the predicted emotion label alongside a bar chart showing per-class softmax probabilities.

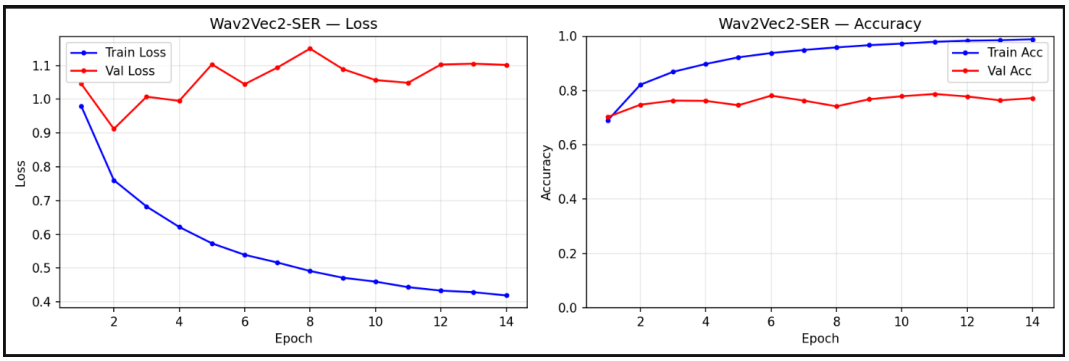


**Figure 1:** Training and validation accuracy curves during Wav2Vec2-base fine-tuning over 14 epochs until early stopping. The validation accuracy plateaus and the best checkpoint is saved at peak validation performance.

# 4. Results

## 4.1 Overall Model Comparison

Table 2 presents test set accuracy and Macro F1 scores for all trained models evaluated on the speaker-independent held-out test set (1,199 clips, 18 actors unseen during training).

**Table 2:** Test set performance across all model architectures on the speaker-independent held-out test set ($n = 1{,}199$ clips, 18 actors).

| Model | Architecture | Test Accuracy | Macro F1 |
|---|---|---|---|
| Model 1 | SVM on 260-dim scalar features | 52.6% | 0.517 |
| Model 1 | Random Forest on 260-dim scalar features | 52.0% | 0.513 |
| Model 2 | CNN + BiLSTM + Attention (Mel spectrogram) | 67.4% | 0.673 |
| Model 3 | EfficientNet-B0 (3-channel spectrogram) | 67.5% | 0.673 |
| Model 4 | ResNet-18 Dual-Input (Mel + waveform) | 62.1% | 0.619 |
| Model 5 | Multi-Feature CNN + BiLSTM (267-dim joint) | 70.7% | 0.705 |
| **Model 6** | **Wav2Vec2-base fine-tuned** | **75.3%** | **0.749** |
| Ensemble | SLSQP-optimized weights + TTA | 75.2% | 0.751 |

The progression from SVM (52.6%) through Multi-Feature CNN + BiLSTM (70.7%) to Wav2Vec2 (75.3%) illustrates the benefit of richer feature representations and pretraining on large unlabeled corpora. Notably, the SLSQP ensemble nearly matched Wav2Vec2 individually, reflecting that ensemble diversity was limited when one model dominated performance.

## 4.2 Per-Emotion Performance (Wav2Vec2)

Table 3 reports per-class precision, recall, and F1-score for the best-performing model on the held-out test set.

**Table 3:** Per-class classification report for Wav2Vec2-base on the speaker-independent test set ($n = 1{,}199$ clips). Bold row indicates macro average across all classes.

| Emotion | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Angry | 0.828 | 0.888 | 0.857 | 250 |
| Fear | 0.638 | 0.840 | 0.725 | 250 |
| Happy | 0.854 | 0.656 | 0.742 | 250 |
| Neutral | 0.736 | 0.910 | 0.813 | 199 |
| Sad | 0.768 | 0.504 | 0.609 | 250 |
| **Macro avg** | **0.765** | **0.760** | **0.749** | **1,199** |

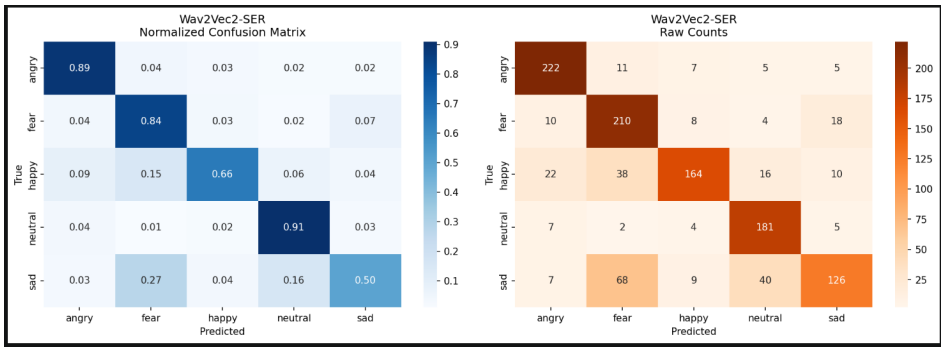


**Figure 2:** Confusion matrix for Wav2Vec2-base on the speaker-independent test set (1,199 clips, 18 actors). Rows represent true labels; columns represent predicted labels. The primary confusion occurs between Sad and Neutral, two acoustically similar low-arousal states.

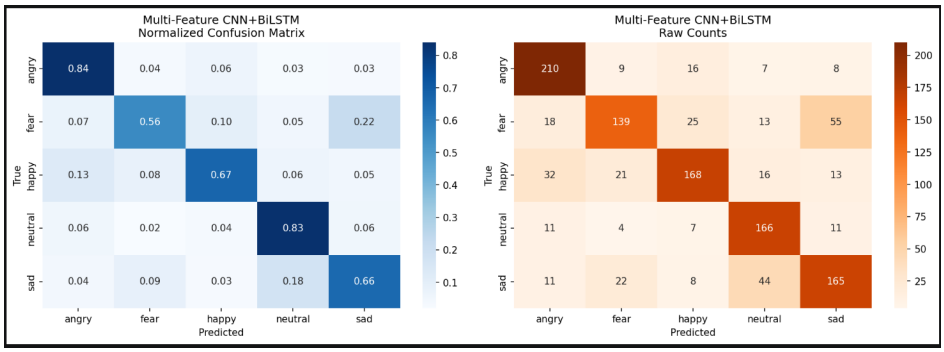


**Figure 3:** Confusion matrix for the Multi-Feature CNN + BiLSTM (Model 5) on the same speaker-independent test set, shown for architectural comparison. The Sad - Neutral confusion is more pronounced in this model (70.7%) than in Wav2Vec2 (75.3%).

The strongest result was for Angry (F1: 0.857), which benefits from high-arousal acoustic cues: elevated pitch, increased energy, faster speech rate, that are reliably present and distinctive in acted speech. Neutral also achieved high recall (0.910), likely because neutral speech provides a stable acoustic baseline that is consistently identified. Sad exhibited the weakest recall (0.504), meaning the model correctly identified only approximately half of sad utterances, with the remainder most commonly misclassified as Neutral. This reflects the genuine acoustic similarity between sadness and neutrality: both are low-arousal states with similar energy levels and speech rates, separated only by subtle timbral differences that even trained human raters find difficult to label consistently.

# 5. Discussion

## 5.1 Interpretation of Results

The primary finding, that fine-tuning a self-supervised transformer pretrained on 960 hours of unlabeled speech substantially outperforms architectures trained from scratch on 24,500 labeled examples, aligns with results reported in recent SER literature. Pepino et al. (2021) demonstrated that Wav2Vec2-base embeddings achieve competitive SER performance with minimal fine-tuning, attributing the advantage to the model's ability to learn phoneme-level and prosodic representations during pretraining that are directly relevant to emotion. The present results extend this finding to a more rigorous evaluation protocol: the 75.3% test accuracy reported here was achieved under strict speaker independence across two combined datasets, whereas many published comparisons evaluate on single datasets with relaxed speaker controls.

The Multi-Feature CNN + BiLSTM (Model 5) achieved the strongest result (70.7%) among architectures trained from scratch, confirming the hypothesis that jointly learning cross-feature interactions rather than training separate models on individual feature types provides a measurable advantage. The gap between Model 5 (70.7%) and Wav2Vec2 (75.3%) illustrates the ceiling imposed by the amount of labeled data available relative to model capacity.

The underperformance of the ResNet-18 dual-input model (62.1%) relative to EfficientNet-B0 (67.5%) and CNN + BiLSTM (67.4%), despite its architectural complexity, is consistent with the observation that larger models require more data to train effectively. ResNet-18's waveform branch processes raw samples at 8 kHz through 1D convolutions, a promising design that likely benefits from more data than the approximately 4,900 original training clips provide.

## 5.2 The Speaker Independence Advantage and Its Cost

The 75.3% accuracy achieved here stands in contrast to published results on RAVDESS alone, where speaker-dependent systems regularly report 90 - 95% (Schuller, 2018). This gap is not a failure of the present approach; it is the expected consequence of a more honest evaluation. A system that achieves 90% accuracy on RAVDESS under speaker-dependent conditions has learned to associate voices with emotion labels, a task that collapses at deployment when

it encounters an unfamiliar speaker. A system achieving 75% under speaker-independent conditions has learned something meaningfully different: acoustic patterns of emotion that hold across speakers, ages, genders, and accents.

This distinction has real consequences for deployed systems. In any real application: clinical monitoring, interactive assistants, call center analysis, the model will encounter speakers it has never processed before. Only speaker-independent evaluation provides a reliable estimate of how the system will actually perform.

### 5.3 Limitations

Several limitations constrain the generalizability of these findings. First, both RAVDESS and CREMA-D consist of *acted* emotional speech, professional actors performing scripted utterances designed to express specific emotions clearly. Natural, spontaneous emotional speech (as occurs in conversation, clinical interviews, or phone calls) is acoustically more ambiguous, more variable, and more likely to mix emotional states. Models trained exclusively on acted speech may not generalize to naturalistic data without additional domain adaptation.

Second, the five-class schema used here (angry, fear, happy, neutral, sad) is a simplification of the continuous, multidimensional nature of human emotion. Alternative frameworks, such as dimensional models specifying valence and arousal continuously, may better capture the gradations between emotional states that posed difficulties for the classifier, particularly the Neutral - Sad boundary.

Third, the dataset is demographically constrained: RAVDESS actors are predominantly professional North American English speakers recorded in studio conditions, and while CREMA-D adds diversity in age, ethnicity, and accent, both corpora over-represent certain speaker populations. A system trained on this data may perform unevenly on speakers from underrepresented linguistic or cultural backgrounds.

Finally, the hardware constraint of a 4 GB VRAM GPU limited Wav2Vec2 training to a batch size of 8, which may have reduced convergence stability. Access to larger GPU resources could support larger batch sizes and potentially higher accuracy.

### 5.4 Future Work

Several extensions would strengthen and expand this work. Cross-corpus evaluation: training on one dataset and testing on the other, would provide an even stricter test of generalization than speaker-independent splits within a combined corpus. Fine-tuning on naturalistic conversational speech (e.g., MSP-Podcast, IEMOCAP) following pretraining on acted data represents a domain adaptation approach aligned with real-world deployment needs.

Dimensionality reduction and visualization of the Wav2Vec2 representations (e.g., via t-SNE or UMAP) would provide interpretable insight into what the model has learned, whether emotional clusters are well-separated in the representation space and which emotions most overlap.

From an engineering perspective, model compression (quantization, knowledge distillation) would reduce inference latency and memory footprint for deployment in resource-constrained environments such as mobile applications, enabling broader real-world impact.

## 6. Culmination of Skills and Contribution to the Field

This project represents the synthesis of skills developed across the dual-degree program in Mathematics (Applied Statistics) and Data Science at UMass Dartmouth, brought together in a single end-to-end system.

From the **mathematics and statistics** curriculum: the design of the speaker-independent evaluation framework is a direct application of statistical methodology, understanding sampling, generalization, and the distinction between in-sample performance and out-of-sample validity. The ensemble weight optimization using SLSQP is an application of constrained numerical optimization. The analysis of per-class precision, recall, and F1 reflects statistical reasoning about Type I and Type II errors in multi-class classification problems.

From the **data science and machine learning** curriculum: the feature extraction pipeline required implementing signal processing transforms (STFT, Mel filterbanks, phase vocoder) in code; the model training required understanding backpropagation, optimization, regularization, and transfer learning at a level of depth that goes beyond course-level exposure; and the deployment required integrating a trained model into a web-serving application accessible to users with no technical background.

The project also required skills that no single course explicitly teaches: debugging a 15-step pipeline across twelve source files, maintaining correctness invariants (the leakage assertion) across a large codebase, making informed architectural decisions under hardware constraints, and communicating technical results clearly to a non-specialist audience.

From the perspective of contribution to the field, this project makes three modest but genuine contributions. First, it provides a publicly accessible implementation of speaker-independent SER evaluation that other students and practitioners can use as a reference baseline. Second, it combines RAVDESS and CREMA-D at the actor-split level, a combination that appears infrequently in published work relative to single-dataset evaluations, and provides concrete accuracy numbers under this more demanding protocol. Third, the live web application demonstrates that research-quality SER is achievable with consumer-grade hardware and open-source tools, lowering the barrier for practitioners who wish to incorporate emotion recognition into applied systems.

For the student personally, this project provided the experience of building something that exists beyond the classroom: a functioning system that anyone with an internet connection

can interact with, demonstrating a complete cycle from scientific question to deployed product. This is the standard expected in industry and graduate research alike, and it is what this project was designed to achieve.

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